

## AD 2 AERODROMES

## RJSC AD 2.1 AERODROME LOCATION INDICATOR AND NAME

## RJSC - YAMAGATA

## RJSC AD 2.2 AERODROME GEOGRAPHICAL AND ADMINISTRATIVE DATA

|   |                                                                                                    |                                                                                                                                                                                                                                                                                 |
|---|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | ARP coordinates and site at AD                                                                     | 382443N 1402216E<br>006°/ 1.0km from RWY 01 THR                                                                                                                                                                                                                                 |
| 2 | Direction and distance from (city)                                                                 | 10.2nm NNE from Yamagata city                                                                                                                                                                                                                                                   |
| 3 | Elevation/ Reference temperature                                                                   | 345ft / 30 °C (2004-2008)                                                                                                                                                                                                                                                       |
| 4 | Geoid undulation at AD ELEV<br>PSN                                                                 | 135ft                                                                                                                                                                                                                                                                           |
| 5 | MAG VAR/ Annual change                                                                             | 9°W (2024) /4°W                                                                                                                                                                                                                                                                 |
| 6 | AD Administration, address,<br>telephone, telefax, telex, AFS,<br>e-mail and/or Web-site addresses | Yamagata Airport Office(Yamagata Pref)<br>3008 Kashiwabarashinrin Hanyuu Higashine-shi Yamagata 999-3776 Japan<br>Tel: 0237-48-1313 Fax: 0237-48-1659<br>e-mail:yyamakuko@pref.yamagata.jp<br>Web-site: <a href="http://www.pref.yamagata.jp/">http://www.pref.yamagata.jp/</a> |
| 7 | Types of traffic permitted (IFR/<br>VFR)                                                           | IFR/VFR                                                                                                                                                                                                                                                                         |
| 8 | Remarks                                                                                            | Yamagata Airport Branch(CAB)<br>3008, Shinrin, Aza-Kashiwabara, Oaza-Hanyu, Higashine-shi, Yamagata Pref.<br>Tel: 0237-48-1118                                                                                                                                                  |

## RJSC AD 2.3 OPERATIONAL HOURS

|    |                           |                                                                                                                      |
|----|---------------------------|----------------------------------------------------------------------------------------------------------------------|
| 1  | AD Administration         | 2300 - 1100                                                                                                          |
| 2  | Customs and immigration   | On request<br>Customs: 023-641-0504<br>Immigration: 0234-22-2746                                                     |
| 3  | Health and sanitation     | On request<br>Quarantine(human): 022-367-8101<br>Quarantine(animal): 022-383-2302<br>Quarantine(plant): 022-362-6916 |
| 4  | AIS Briefing Office       | Nil                                                                                                                  |
| 5  | ATS Reporting Office(ARO) | Nil                                                                                                                  |
| 6  | MET Briefing Office       | H24 (TOKYO)                                                                                                          |
| 7  | ATS                       | 2300 - 1100                                                                                                          |
| 8  | Fuelling                  | 2300 - 1100                                                                                                          |
| 9  | Handling                  | 2300 - 1100                                                                                                          |
| 10 | Security                  | 2300 - 1100                                                                                                          |
| 11 | De-icing                  | Nil                                                                                                                  |
| 12 | Remarks                   | Nil                                                                                                                  |

**RJSC AD 2.4 HANDLING SERVICES AND FACILITIES**

|   |                                         |                                                                                           |
|---|-----------------------------------------|-------------------------------------------------------------------------------------------|
| 1 | Cargo-handling facilities               | 2t-forklift car x 1, 7m-belt loading car x 2, Available up to only bulk loading aircraft. |
| 2 | Fuel/ oil types                         | JET A-1                                                                                   |
| 3 | Fuelling facilities/ capacity           | Fuel truck / 200kl(JET A-1)                                                               |
| 4 | De-icing facilities                     | De-icing car                                                                              |
| 5 | Hangar space for visiting aircraft      | Nil                                                                                       |
| 6 | Repair facilities for visiting aircraft | Nil                                                                                       |
| 7 | Remarks                                 | Nil                                                                                       |

**RJSC AD 2.5 PASSENGER FACILITIES**

|   |                      |                                                       |
|---|----------------------|-------------------------------------------------------|
| 1 | Hotels               | around Airport                                        |
| 2 | Restaurants          | At Airport                                            |
| 3 | Transportation       | Airport Shuttle(Reservation required) and Taxi        |
| 4 | Medical facilities   | Hospital in Higashine city 6km                        |
| 5 | Bank and Post Office | Bank in Higashine city, Post Office in Higashine city |
| 6 | Tourist Office       | Tourist Office in Higashine city                      |
| 7 | Remarks              | Nil                                                   |

**RJSC AD 2.6 RESCUE AND FIRE FIGHTING SERVICES**

|   |                                             |                                  |
|---|---------------------------------------------|----------------------------------|
| 1 | AD category for fire fighting               | CAT 7                            |
| 2 | Rescue equipment                            | Chemical fire fighting truck x 2 |
| 3 | Capability for removal of disabled aircraft | A300,B767                        |
| 4 | Remarks                                     | Nil                              |

**RJSC AD 2.7 SEASONAL AVAILABILITY-CLEARING**

|   |                             |                                                                |
|---|-----------------------------|----------------------------------------------------------------|
| 1 | Types of clearing equipment | Snow remove equipments:<br>truck x 10 , rotary x 3 , dozer x 1 |
| 2 | Clearance priorities        | RWY , TWY , Apron (The same)                                   |
| 3 | Remarks                     | Nil                                                            |

## RJSC AD 2.8 APRONS, TAXIWAYS AND CHECK LOCATIONS DATA

|   |                                     |                                                                                                                                                       |
|---|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Apron surface and strength          | Apron :<br>Surface:cement-concrete, Strength: PCR 750/R/C/W/T<br>Small ACFT Apron:<br>Surface:asphalt-concrete, Strength: AUW5,700kg/0.28MPa          |
| 2 | Taxiway width, surface and strength | Width:30m<br>Surface:asphalt-concrete<br>Strength: PCR 704/F/B/X/T                                                                                    |
| 3 | ACL and elevation                   | Not available                                                                                                                                         |
| 4 | VOR checkpoints                     | Not available                                                                                                                                         |
| 5 | INS checkpoints                     | Spot NR<br>1:382435.52N 1402201.11E<br>1-1:382436.27N 1402201.52E<br>2:382437.31N 1402201.73E<br>3:382439.26N 1402201.81E<br>5:382441.02N 1402201.68E |
| 6 | Remarks                             | Nil                                                                                                                                                   |

## RJSC AD 2.9 SURFACE MOVEMENT GUIDANCE AND CONTROL SYSTEM AND MARKINGS

|   |                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                         |
|---|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Use of aircraft stand ID signs, TWY guide lines and Visual docking/ parking guidance system of aircraft stands | Nil                                                                                                                                                                                                                                                                                                                                                                     |
| 2 | RWY and TWY markings and LGT                                                                                   | RWY:01/19<br>(Marking) RWY designation, RWY CL, RWY THR, TDZ, RWY side stripe, Aiming point, RWY turn pad edge, RWY turn pad CL, RWY turn pad aiming (LGT) RCLL, REDL, RTHL, RENL, RTZL(FOR RWY01), WBAR(FOR RWY01), RWY DIST marker LGT, Turning point indicator LGT<br>TWY:<br>(Marking) TWY CL, RWY HLDG PSN, TWY side stripe marking (LGT) TWY edge LGT, TWY CL LGT |
| 3 | Stop bars                                                                                                      | Nil                                                                                                                                                                                                                                                                                                                                                                     |
| 4 | Remarks                                                                                                        | (Marking) Overrun area<br>(LGT) Apron flood LGT                                                                                                                                                                                                                                                                                                                         |

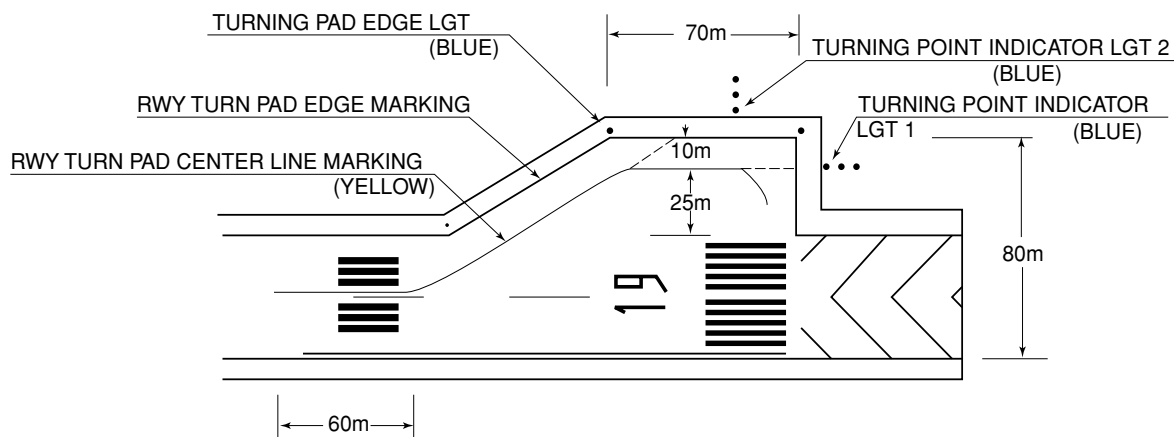
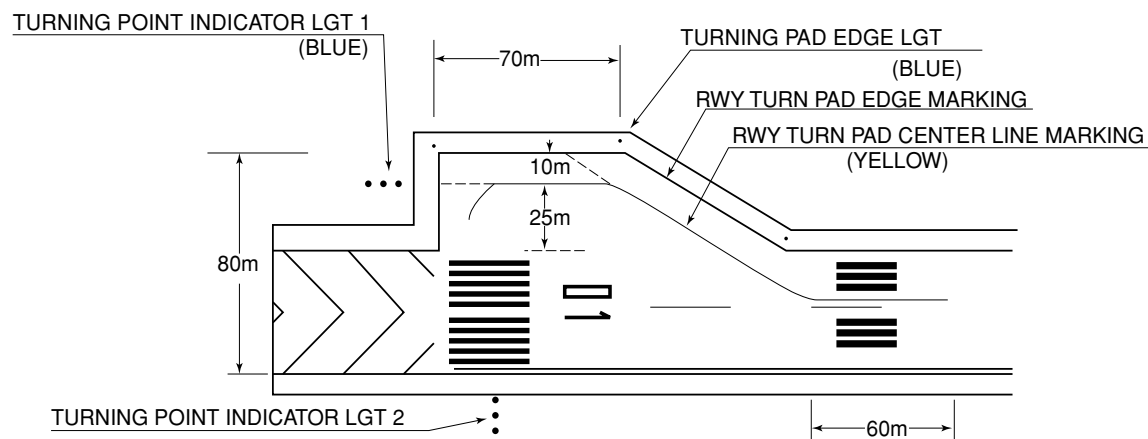
## 180° turn on RWY

B-767型機の滑走路180° 転回実施要領

1. 滑走路中心線からターニングパッド中心線標識に従って進行する。
2. 転回灯1が一直線に見えるように進行し、転回灯2が一直線に見えた時転回を開始する。転回時はMAX STEERING ANGLEを使用する。

Procedure of 180° turn on RWY for B-767 aircraft

1. Proceed along the RWY Turn Pad Center Line Marking.
  2. Proceed along the RWY Turn Pad Center Line Marking to see the Turning Point Indicator Light 1 on a straight line, then commence turn at the spot where you (pilot) can see the Turning Point Indicator Light 2 on a straight line at an angle of 9 o'clock.
- When turning, take MAX STEERING ANGLE.

YAMAGATA AP

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**RJSC AD 2.10 AERODROME OBSTACLES**

■ In Area2 See Obstacle data

■ In Area3 To be developed

## RJSC AD 2.11 METEOROLOGICAL INFORMATION PROVIDED

|    |                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Associated MET Office                                                  | TOKYO                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 2  | Hours of service<br>MET Office outside hours                           | H24 (TOKYO)                                                                                                                                                                                                                                                                                                                                                                                                             |
| 3  | Office responsible for TAF preparation<br>Periods of validity          | Nil                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 4  | Trend forecast<br>Interval of issuance                                 | Nil                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5  | Briefing/ consultation provided                                        | Briefing is available upon inquiry at TOKYO                                                                                                                                                                                                                                                                                                                                                                             |
| 6  | Flight documentation<br>Language(s) used                               | C<br>En                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 7  | Charts and other information available<br>for briefing or consultation | S <sub>6</sub> , U <sub>85</sub> , U <sub>7</sub> , U <sub>5</sub> , U <sub>3</sub> , U <sub>25</sub> , U <sub>2</sub> /T <sub>r</sub> , P <sub>S</sub> , P <sub>5</sub> , P <sub>3</sub> , P <sub>25</sub> , P <sub>SWE</sub> , P <sub>SWF</sub> , P <sub>SWG</sub> , P <sub>SWI</sub> , P <sub>SWM</sub> , P <sub>SW</sub> (domestic), E, C, W <sub>E</sub> , W <sub>F</sub> , W <sub>G</sub> , W <sub>I</sub> , W, N |
| 8  | Supplementary equipment<br>available for providing information         | Nil                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 9  | ATS units provided with information                                    | RADIO                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 10 | Additional information(limitation of service, etc.)                    | Nil                                                                                                                                                                                                                                                                                                                                                                                                                     |

## RJSC AD 2.12 RUNWAY PHYSICAL CHARACTERISTICS

| Designations<br>RWY NR | TRUE BRG               | Dimensions of<br>RWY(M) | Strength(PCR) and<br>surface of RWY                                                          | THR coordinates<br>THR geoid undulation | THR elevation and<br>highest elevation of TDZ<br>of precision APP RWY |
|------------------------|------------------------|-------------------------|----------------------------------------------------------------------------------------------|-----------------------------------------|-----------------------------------------------------------------------|
| 1                      | 2                      | 3                       | 4                                                                                            | 5                                       | 6                                                                     |
| 01                     | 006.57°                | 2000×45                 | PCR<br>1013/F/D/X/T<br>Asphalt Concrete                                                      | 382410.61N<br>1402212.06E<br>135.4FT    | THR ELEV: 346.9ft<br>TDZ ELEV: 346.9ft                                |
| 19                     | 186.57°                | 2000×45                 | PCR<br>1013/F/D/X/T<br>Asphalt Concrete                                                      | 382515.04N<br>1402221.52E<br>135.4FT    | THR ELEV: 353ft                                                       |
| Slope of RWY           | Strip<br>Dimensions(M) |                         | RESA (Overrun)<br>Dimensions(M)                                                              |                                         | Remarks                                                               |
| 7                      | 10                     |                         | 11                                                                                           |                                         | 14                                                                    |
| See AD2.24 AD chart    | 2120×300<br>2120×300   |                         | 152 × (MNM:157 MAX:300)*<br>90 × (MNM:90 MAX:300)*<br>*For detail, ask airport administrator |                                         | RWY grooving : 2000m×30m                                              |

## RJSC AD 2.13 DECLARED DISTANCES

| RWY Designator | TORA<br>(m) | TODA<br>(m) | ASDA<br>(m) | LDA<br>(m) | Remarks |
|----------------|-------------|-------------|-------------|------------|---------|
| 1              | 2           | 3           | 4           | 5          | 6       |
| 01             | 2000        | 2000        | 2000        | 2000       | Nil     |
| 19             | 2000        | 2000        | 2000        | 2000       | Nil     |

## RJSC AD 2.14 APPROACH AND RUNWAY LIGHTING

| RWY Designator                                                                                                                 | APCH<br>LGT<br>type<br>LEN<br>INTST | RTHL<br>Color<br>WBAR | PAPI<br>(VASIS)<br>Angle<br>DIST FM<br>THR<br>MEHT | RTZL<br>LEN | RCLL<br>LEN<br>Spacing<br>Color<br>INTST          | REDL<br>LEN<br>Spacing<br>Color<br>INTST             | RENL<br>Color<br>WBAR | STWL<br>LEN<br>Color |
|--------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|-----------------------|----------------------------------------------------|-------------|---------------------------------------------------|------------------------------------------------------|-----------------------|----------------------|
| 1                                                                                                                              | 2                                   | 3                     | 4                                                  | 5           | 6                                                 | 7                                                    | 8                     | 9                    |
| 01                                                                                                                             | PALS<br>(CAT I)<br>900m<br>LIH      | Green<br>Green        | PAPI<br>3.0° /Left<br>410m<br>61ft                 | 900m        | 2000m<br>30m<br>Coded color<br>(White/Red)<br>LIH | 2000m<br>60m<br>Coded color<br>(White/Yellow)<br>LIH | Red                   | Nil (*2)             |
| 19                                                                                                                             | SALS<br>(*1)<br>420m<br>LIH         | Green<br>Nil          | PAPI<br>3.0°/Left<br>386.7m<br>61ft                | Nil         | 2000m<br>30m<br>Coded color<br>(White/Red)<br>LIH | 2000m<br>60m<br>Coded color<br>(White/Yellow)<br>LIH | Red                   | Nil (*2)             |
| Remarks                                                                                                                        |                                     |                       |                                                    |             |                                                   |                                                      |                       |                      |
| 10                                                                                                                             |                                     |                       |                                                    |             |                                                   |                                                      |                       |                      |
| SALS with APCH LGT beacon (600m and 900m FM RWY THR ) (*1)<br>Overrun area edge LGT(LEN:60m, color:Red) (*2)<br>CGL for RWY 19 |                                     |                       |                                                    |             |                                                   |                                                      |                       |                      |

## RJSC AD 2.15 OTHER LIGHTING, SECONDARY POWER SUPPLY

|   |                                                          |                                                                                                                                          |
|---|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | ABN/IBN location, characteristics and hours of operation | ABN : 382445N/1402156E, White/Green EV4.3sec, HO                                                                                         |
| 2 | LDI location and LGT<br>Anemometer location and LGT      | LDI : Nil<br>Anemometer :<br>RWY01 : On the north side of 170m FM RWY01 THR, LGTD<br>RWY19 : On the north side of 30m FM RWY19 THR, LGTD |
| 3 | TWY edge and center line lighting                        | TWY edge and center line lights installed, see AD 2.9                                                                                    |
| 4 | Secondary power supply/switch-over time                  | Within 1sec : REDL, RCLL, RTHL, RENL, WBAR, Turning point indicator LGT, Overrun area edge LGT<br>Within 15sec : Other LGT               |
| 5 | Remarks                                                  | WDI LGT                                                                                                                                  |

**RJSC AD 2.16 HELICOPTER LANDING AREA**

|     |
|-----|
| Nil |
|-----|

**RJSC AD 2.17 ATS AIRSPACE**

| Designation and lateral limits |                                         | Vertical limits (ft) | Airspace classification | ATS unit call sign Language | Remarks |
|--------------------------------|-----------------------------------------|----------------------|-------------------------|-----------------------------|---------|
| 1                              |                                         | 2                    | 3                       | 4                           | 6       |
| Yamagata Information zone      | Area within a radius of 9km(5NM) of ARP | 3000 or below        | E                       | Yamagata radio En           |         |

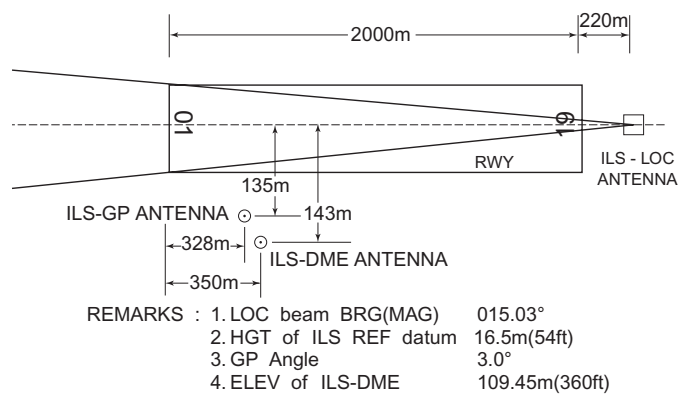
**RJSC AD 2.18 ATS COMMUNICATION FACILITIES**

| Service designation | Call sign      | Frequency                              | Hours of operation | Remarks    |
|---------------------|----------------|----------------------------------------|--------------------|------------|
| 1                   | 2              | 3                                      | 4                  | 5          |
| AFIS                | Yamagata Radio | 122.7MHz(1)<br>126.2MHz<br>243.0MHz(E) | 2300 - 1100        | (1)Primary |

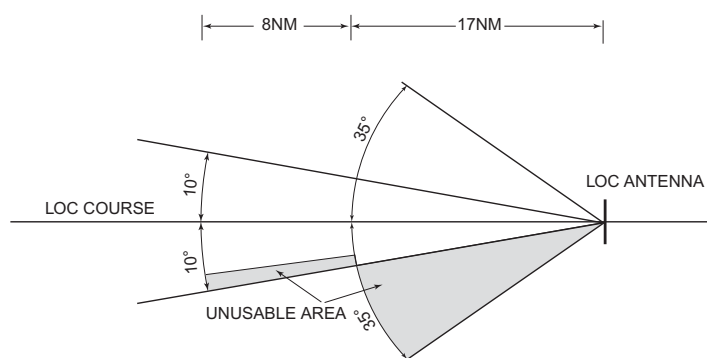


## RJSC AD 2.19 RADIO NAVIGATION AND LANDING AIDS

| Type of aid<br>(VOR<br>declination) | ID  | Frequency           | Hours of<br>operation | Position of<br>transmitting<br>antenna<br>coordinates | Elevation<br>of<br>DME<br>transmit-<br>ting<br>antenna | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------|-----|---------------------|-----------------------|-------------------------------------------------------|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1                                   | 2   | 3                   | 4                     | 5                                                     | 6                                                      | 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| VOR<br>(8°W/2016)                   | YTE | 113MHz              | H24                   | 382319.04N<br>1402128.63E                             |                                                        | VOR/DME Unusable:<br>030°-040° beyond 35nm BLW 8000ft.<br>040°-060° beyond 25nm BLW 8000ft.<br>060°-070° beyond 15nm BLW 8000ft.<br>070°-080° beyond 15nm BLW 7000ft.<br>080°-110° beyond 20nm BLW 7000ft.<br>110°-150° beyond 15nm BLW 7000ft.<br>150°-160° beyond 25nm BLW 9000ft.<br>160°-170° beyond 30nm BLW 9000ft.<br>170°-180° beyond 25nm BLW 9000ft.<br>250°-260° beyond 35nm BLW 9000ft.<br>260°-270° beyond 35nm BLW 8000ft.<br>300°-320° beyond 30nm BLW 9000ft.<br>320°-330° beyond 20nm BLW 9000ft.<br>330°-340° beyond 30nm BLW 10000ft. |
| DME                                 | YTE | 1164MHz<br>(CH-77X) | H24                   | 382319.04N<br>1402128.63E                             | 361ft                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| ILS-LOC 01                          | IYT | 110.1MHz            | 2300 - 1100           | 382522.13N<br>1402222.56E                             |                                                        | LOC: 220m(722ft) away FM RWY 19<br>THR, BRG(MAG) 015.03°<br>LOC Unusable in the following area<br>(1)beyond 10 degrees<br>East side of LOC course.<br>(2)beyond 8 degrees<br>East side of LOC course<br>beyond 17NM fm LOC antenna.                                                                                                                                                                                                                                                                                                                      |
| ILS-GP 01                           | -   | 334.4MHz            | 2300 - 1100           | 382420.66N<br>1402219.20E                             |                                                        | GP : 328m(1076ft) inside FM RWY 01<br>THR, 135m (443ft) E of RCL, angle<br>3.0°HGT of ILS Ref datum 16.5m(54ft).                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| ILS-DME 01                          | IYT | 999MHz<br>(CH-38X)  | 2300 - 1100           | 382421.31N<br>1402219.57E                             | 360ft                                                  | DME :350m(1148ft)inside FM RWY<br>01,143m(469ft) E of RCL.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| MSAS                                |     | 1575.42MHz          | H24                   |                                                       |                                                        | Transmitting antennas are<br>satellite based.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

ILS

LOC unusable is the following area : (1) beyond 10° East side of LOC course.  
 (2) beyond 8° East side of LOC course beyond 17nm from LOC ANT.

**RJSC AD 2.20 LOCAL TRAFFIC REGULATIONS**

## 1. Airport regulations

On use of YAMAGATA airport, aircraft operator is required to notify the airport administrator in advance.

## 2. Taxiing to and from stands

Nil

## 3. Parking area for small aircraft(General aviation)

Nil

## 4. Parking area for helicopters

Nil

## 5. Apron - taxiing during winter conditions

Nil

## 6. Taxiing - limitations

Nil

## 7. School and training flights - technical test flights - use of runways

Nil

## 8. Helicopter traffic - limitation

Nil

## 9. Removal of disabled aircraft from runways

Nil

## RJSC AD 2.21 NOISE ABATEMENT PROCEDURES

Nil

## RJSC AD 2.22 FLIGHT PROCEDURES

## 1. TAKE OFF MINIMA

|                                                    | RWY | ACFT<br>CAT | REDL & RCLL     |     | REDL or RCLL<br>or RCL Marking |     | NIL<br>(DAYTIME ONLY) |     |
|----------------------------------------------------|-----|-------------|-----------------|-----|--------------------------------|-----|-----------------------|-----|
|                                                    |     |             | RVR             | VIS | RVR                            | VIS | RVR                   | VIS |
| Multi-Engine<br>ACFT with<br>TKOF ALTN<br>AP FILED | 01  | A,B,C,D     | 400             | 400 | 400                            | 400 | -                     | 500 |
|                                                    | 19  | A,B,C,D     | -               | 400 | -                              | 400 | -                     | 500 |
| OTHER                                              | 01  | A,B,C,D     | AVBL LDG MINIMA |     |                                |     |                       |     |
|                                                    | 19  | A,B,C,D     |                 |     |                                |     |                       |     |

## 2. OTHER

VFR aircraft intending to land on or fly around the AP, especially south and north of the AP, is recommended to make initial contact with Yamagata RADIO to obtain traffic information at least 15nm far from the AP.

当空港に着陸または空港周辺、特に空港の南及び北側を飛行しようとする VFR の航空機については、交通情報の入手のため、少なくとも 15NM 以遠からの山形 RADIO との通信設定が推奨される。

## RJSC AD 2.23 ADDITIONAL INFORMATION

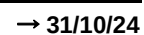
Nil

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RJSC AD 2.24 CHARTS RELATED TO AN AERODROME

Aerodrome/Heliport Chart  
Standard Departure Chart - Instrument (YAMAGATA)  
Standard Departure Chart - Instrument (RUBIS-RNAV)  
Standard Departure Chart - Instrument (NIIGATA-RNAV)  
Standard Departure Chart - Instrument (HANKA-RNAV)  
Standard Arrival Chart - Instrument (YOZAN WEST, YOZAN NORTH-RNAV)  
Standard Arrival Chart - Instrument (MEDET WEST, MEDET SOUTH-RNAV)  
Standard Arrival Chart - Instrument (TUYAH-RNAV)  
Instrument Approach Chart (ILS Z RWY01)  
Instrument Approach Chart (LOC Z RWY01)  
Instrument Approach Chart (ILS Y RWY01)  
Instrument Approach Chart (LOC Y RWY01)  
Instrument Approach Chart (VOR A)  
Instrument Approach Chart (RNP RWY19)  
Other Chart (Visual REP)  
Other Chart (LDG CHART)  
Other Chart (MVA CHART)

## AD CHART



STANDARD DEPARTURE CHART-INSTRUMENT

RJSC / YAMAGATA

SID

YAMAGATA FOUR DEPARTURE

RWY01 : Climb RWY HDG to 2000FT, via YTE R017 to 22.0DME, turn left, via YTE R003 to YTE VOR/DME.

RWY19 : Climb RWY HDG to 2000FT, turn right, direct to YTE VOR/DME, cross YTE VOR/DME at or above 4000FT, via YTE R017 to 15.0DME, turn left, via YTE R354 to YTE VOR/DME.

Note RWY01: 5.0% climb gradient required up to 2000FT.

OBST ALT 1182FT located at 4.2NM 033° FM end of RWY01.

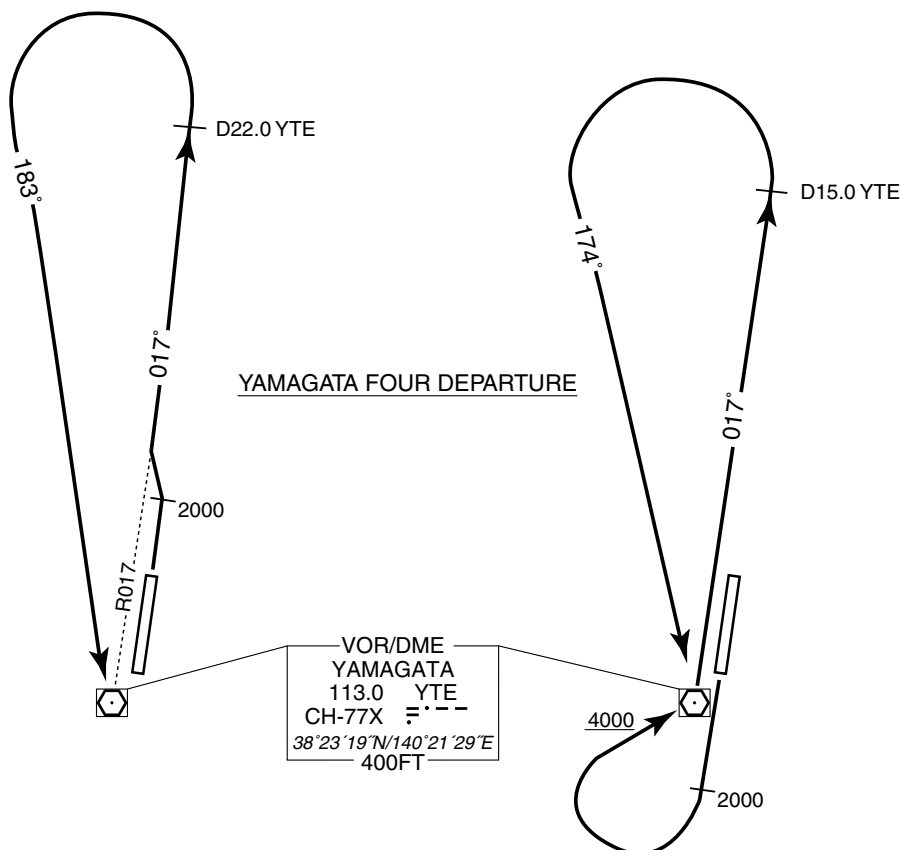
RWY19: 5.0% climb gradient required up to 2000FT.

OBST ALT 788FT located at 3.1NM 181° FM end of RWY19.

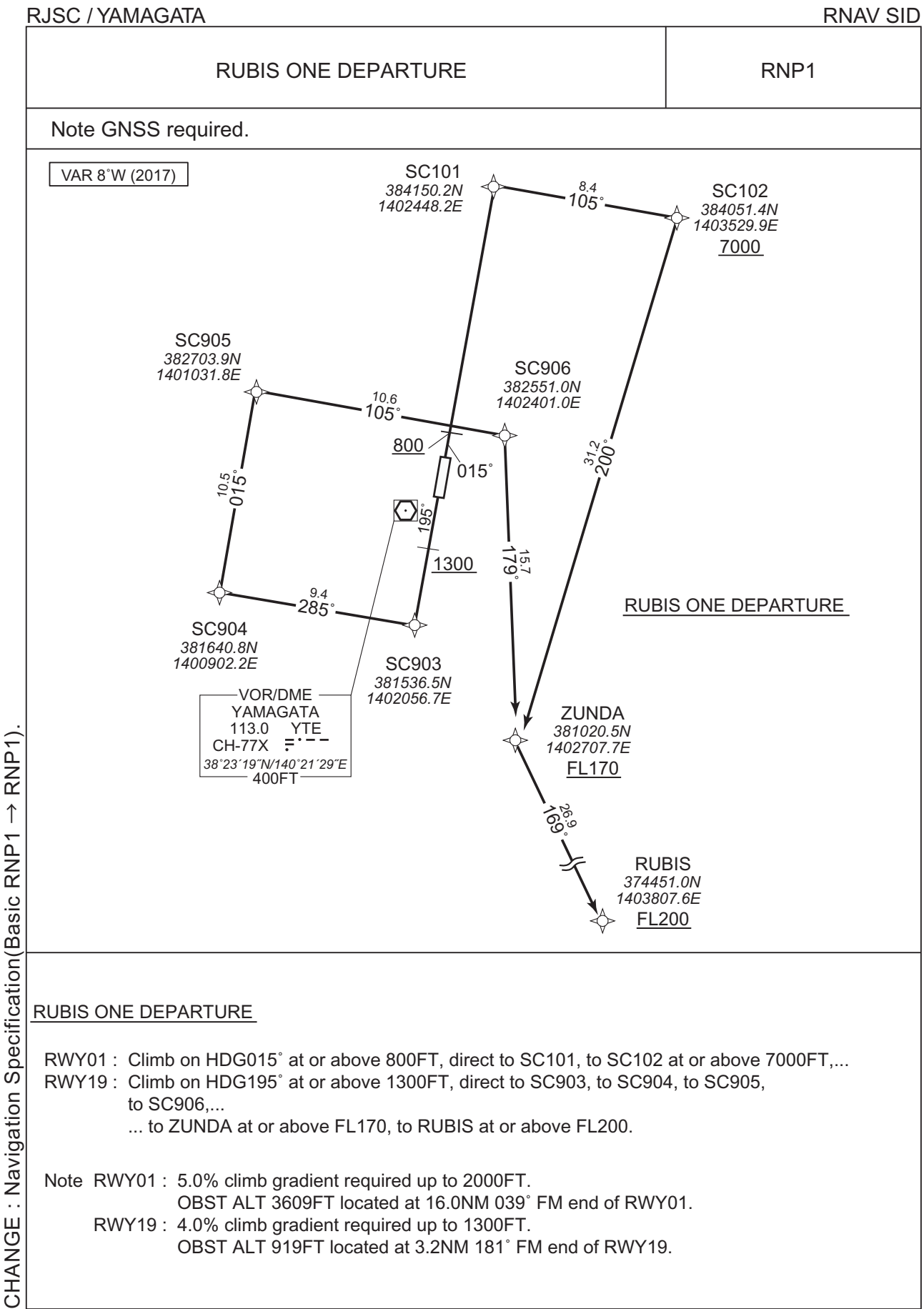
Note No turn before DER.

For RWY01

For RWY19



STANDARD DEPARTURE CHART-INSTRUMENT



CHANGE : Navigation Specification(Basic RNP1 → RNP1).

STANDARD DEPARTURE CHART-INSTRUMENT

RJSC / YAMAGATA

RNAV SID

RUBIS ONE DEPARTURE

RWY01

| Serial Number | Path Descriptor | Waypoint Identifier | Fly Over | Course °M(°T)  | Magnetic Variation | Distance (NM) | Turn Direction | Altitude (FT) | Speed (KIAS) | Vertical Angle | Navigation Specification |
|---------------|-----------------|---------------------|----------|----------------|--------------------|---------------|----------------|---------------|--------------|----------------|--------------------------|
| 001           | VA              | -                   | -        | 015<br>(006.6) | -8.2               | -             | -              | +800          | -            | -              | RNP1                     |
| 002           | DF              | SC101               | -        | -              | -8.2               | -             | -              | -             | -            | -              | RNP1                     |
| 003           | TF              | SC102               | -        | 105<br>(096.6) | -8.2               | 8.4           | -              | +7000         | -            | -              | RNP1                     |
| 004           | TF              | ZUNDA               | -        | 200<br>(192.2) | -8.2               | 31.2          | -              | +FL170        | -            | -              | RNP1                     |
| 005           | TF              | RUBIS               | -        | 169<br>(161.2) | -8.2               | 26.9          | -              | +FL200        | -            | -              | RNP1                     |

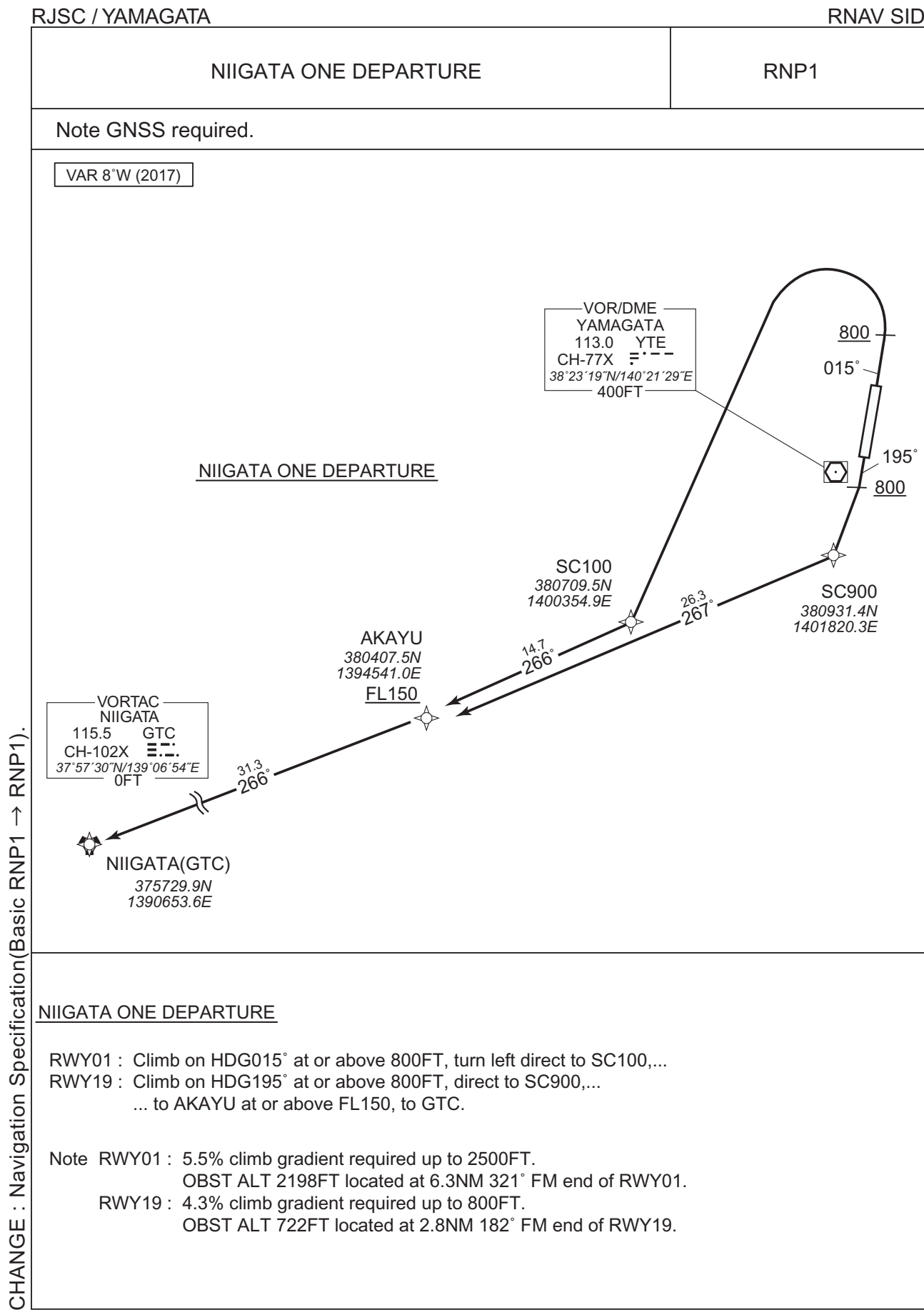
RWY19

| Serial Number | Path Descriptor | Waypoint Identifier | Fly Over | Course °M(°T)  | Magnetic Variation | Distance (NM) | Turn Direction | Altitude (FT) | Speed (KIAS) | Vertical Angle | Navigation Specification |
|---------------|-----------------|---------------------|----------|----------------|--------------------|---------------|----------------|---------------|--------------|----------------|--------------------------|
| 001           | VA              | -                   | -        | 195<br>(186.6) | -8.2               | -             | -              | +1300         | -            | -              | RNP1                     |
| 002           | DF              | SC903               | -        | -              | -8.2               | -             | -              | -             | -            | -              | RNP1                     |
| 003           | TF              | SC904               | -        | 285<br>(276.6) | -8.2               | 9.4           | -              | -             | -            | -              | RNP1                     |
| 004           | TF              | SC905               | -        | 015<br>(006.4) | -8.2               | 10.5          | -              | -             | -            | -              | RNP1                     |
| 005           | TF              | SC906               | -        | 105<br>(096.5) | -8.2               | 10.6          | -              | -             | -            | -              | RNP1                     |
| 006           | TF              | ZUNDA               | -        | 179<br>(171.0) | -8.2               | 15.7          | -              | +FL170        | -            | -              | RNP1                     |
| 007           | TF              | RUBIS               | -        | 169<br>(161.2) | -8.2               | 26.9          | -              | +FL200        | -            | -              | RNP1                     |

CHANGE : Navigation Specification(Basic RNP1 → RNP1).



STANDARD DEPARTURE CHART-INSTRUMENT



STANDARD DEPARTURE CHART-INSTRUMENT

RJSC / YAMAGATA

RNAV SID

NIIGATA ONE DEPARTURE

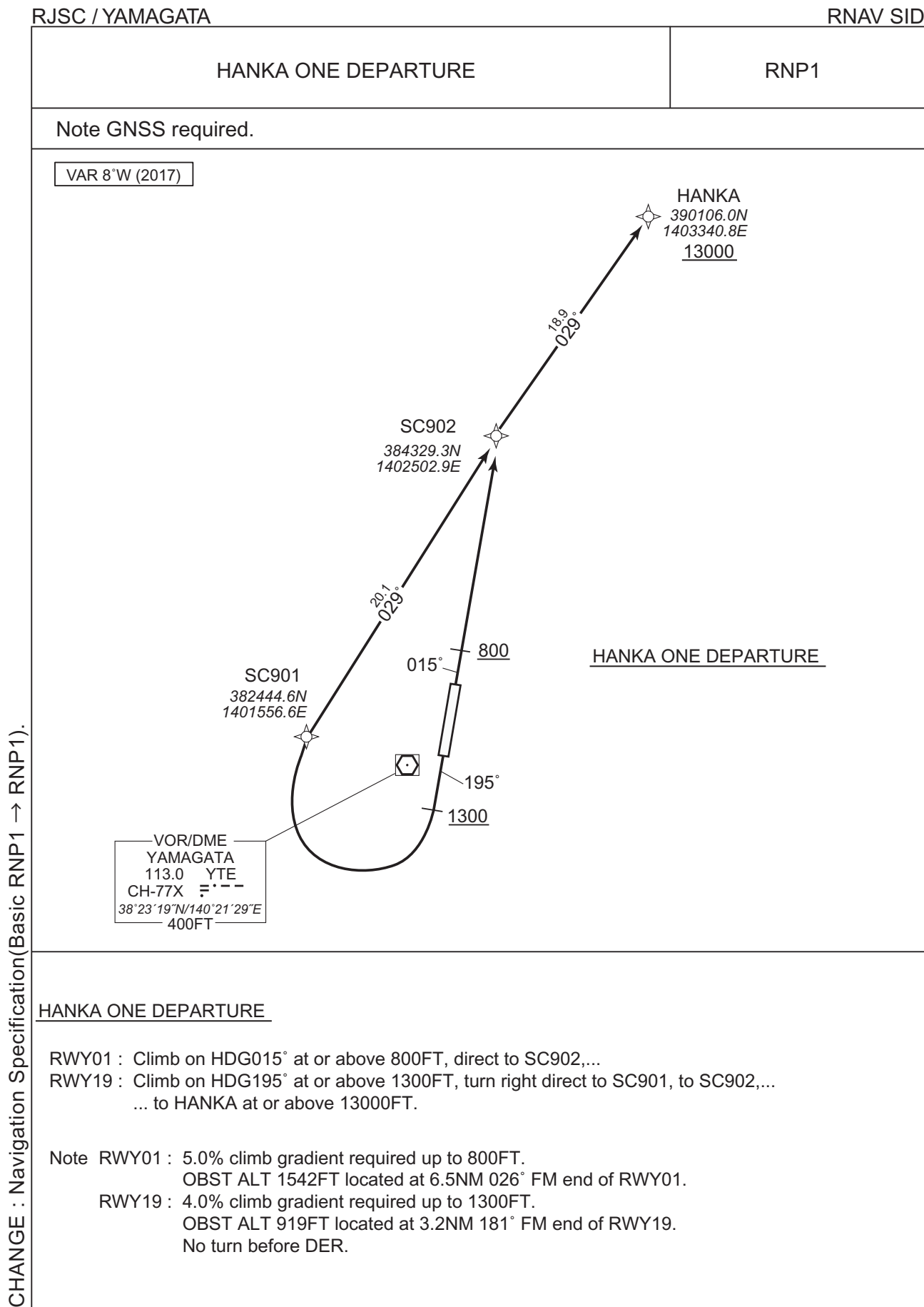
RWY01

| Serial Number | Path Descriptor | Waypoint Identifier | Fly Over | Course °M(°T)  | Magnetic Variation | Distance (NM) | Turn Direction | Altitude (FT) | Speed (KIAS) | Vertical Angle | Navigation Specification |
|---------------|-----------------|---------------------|----------|----------------|--------------------|---------------|----------------|---------------|--------------|----------------|--------------------------|
| 001           | VA              | -                   | -        | 015<br>(006.6) | -8.2               | -             | -              | +800          | -            | -              | RNP1                     |
| 002           | DF              | SC100               | -        | -              | -8.2               | -             | L              | -             | -            | -              | RNP1                     |
| 003           | TF              | AKAYU               | -        | 266<br>(258.2) | -8.2               | 14.7          | -              | +FL150        | -            | -              | RNP1                     |
| 004           | TF              | GTC                 | -        | 266<br>(258.0) | -8.2               | 31.3          | -              | -             | -            | -              | RNP1                     |

RWY19

| Serial Number | Path Descriptor | Waypoint Identifier | Fly Over | Course °M(°T)  | Magnetic Variation | Distance (NM) | Turn Direction | Altitude (FT) | Speed (KIAS) | Vertical Angle | Navigation Specification |
|---------------|-----------------|---------------------|----------|----------------|--------------------|---------------|----------------|---------------|--------------|----------------|--------------------------|
| 001           | VA              | -                   | -        | 195<br>(186.6) | -8.2               | -             | -              | +800          | -            | -              | RNP1                     |
| 002           | DF              | SC900               | -        | -              | -8.2               | -             | -              | -             | -            | -              | RNP1                     |
| 003           | TF              | AKAYU               | -        | 267<br>(258.3) | -8.2               | 26.3          | -              | +FL150        | -            | -              | RNP1                     |
| 004           | TF              | GTC                 | -        | 266<br>(258.0) | -8.2               | 31.3          | -              | -             | -            | -              | RNP1                     |

STANDARD DEPARTURE CHART-INSTRUMENT



CHANGE : Navigation Specification(Basic RNP1 → RNP1).

STANDARD DEPARTURE CHART-INSTRUMENT

RJSC / YAMAGATA RNAV SID

| HANKA ONE DEPARTURE                                   |                 |                     |          |                |                    |               |                |               |              |                |                          |
|-------------------------------------------------------|-----------------|---------------------|----------|----------------|--------------------|---------------|----------------|---------------|--------------|----------------|--------------------------|
| RWY01                                                 |                 |                     |          |                |                    |               |                |               |              |                |                          |
| Serial Number                                         | Path Descriptor | Waypoint Identifier | Fly Over | Course °M(°T)  | Magnetic Variation | Distance (NM) | Turn Direction | Altitude (FT) | Speed (KIAS) | Vertical Angle | Navigation Specification |
| 001                                                   | VA              | -                   | -        | 015<br>(006.6) | -8.2               | -             | -              | +800          | -            | -              | RNP1                     |
| 002                                                   | DF              | SC902               | -        | -              | -8.2               | -             | -              | -             | -            | -              | RNP1                     |
| 003                                                   | TF              | HANKA               | -        | 029<br>(020.8) | -8.2               | 18.9          | -              | +13000        | -            | -              | RNP1                     |
| RWY19                                                 |                 |                     |          |                |                    |               |                |               |              |                |                          |
| Serial Number                                         | Path Descriptor | Waypoint Identifier | Fly Over | Course °M(°T)  | Magnetic Variation | Distance (NM) | Turn Direction | Altitude (FT) | Speed (KIAS) | Vertical Angle | Navigation Specification |
| 001                                                   | VA              | -                   | -        | 195<br>(186.6) | -8.2               | -             | -              | +1300         | -            | -              | RNP1                     |
| 002                                                   | DF              | SC901               | -        | -              | -8.2               | -             | R              | -             | -            | -              | RNP1                     |
| 003                                                   | TF              | SC902               | -        | 029<br>(020.7) | -8.2               | 20.1          | -              | -             | -            | -              | RNP1                     |
| 004                                                   | TF              | HANKA               | -        | 029<br>(020.8) | -8.2               | 18.9          | -              | +13000        | -            | -              | RNP1                     |
| CHANGE : Navigation Specification(Basic RNP1 → RNP1). |                 |                     |          |                |                    |               |                |               |              |                |                          |

**YOZAN WEST ARRIVAL / YOZAN NORTH ARRIVAL**

**RNP1**

Note GNSS required.

VAR 8°W (2017)

**YOZAN WEST ARRIVAL**

**YOZAN NORTH ARRIVAL**

**ELDAK**  
380133.9N  
1394540.4E  
FL150

**SUGGY**  
380424.9N  
1400035.7E  
11000

**YOZAN**  
380617.1N  
1400922.8E  
9000

**IMONY**  
381111.6N  
1401005.0E  
MAX 250KIAS

**YAMAGATA (YTE)**  
382319.0N  
1402128.6E

**VOR/DME YAMAGATA**  
113.0 YTE  
CH-77X  
38°23'19"N/140°21'29"E  
400FT

12.1 085°

7.2 083°

4.9 195°

15.1 225°

CHANGE : Navigation Specification(Basic RNP1 → RNP1).

STANDARD ARRIVAL CHART - INSTRUMENT

RJSC / YAMAGATA

RNAV STAR RWY01

YOZAN WEST ARRIVAL

From ELDAK at or above FL150, to SUGGY at or above 11000FT,  
to YOZAN at or above 9000FT.

| Serial Number | Path Descriptor | Waypoint Identifier | Fly Over | Course °M(°T)  | Magnetic Variation | Distance (NM) | Turn Direction | Altitude (FT) | Speed (KIAS) | Vertical Angle | Navigation Specification |
|---------------|-----------------|---------------------|----------|----------------|--------------------|---------------|----------------|---------------|--------------|----------------|--------------------------|
| 001           | IF              | ELDAK               | -        | -              | -8.2               | -             | -              | +FL150        | -            | -              | RNP1                     |
| 002           | TF              | SUGGY               | -        | 085<br>(076.3) | -8.2               | 12.1          | -              | +11000        | -            | -              | RNP1                     |
| 003           | TF              | YOZAN               | -        | 083<br>(074.8) | -8.2               | 7.2           | -              | +9000         | -            | -              | RNP1                     |

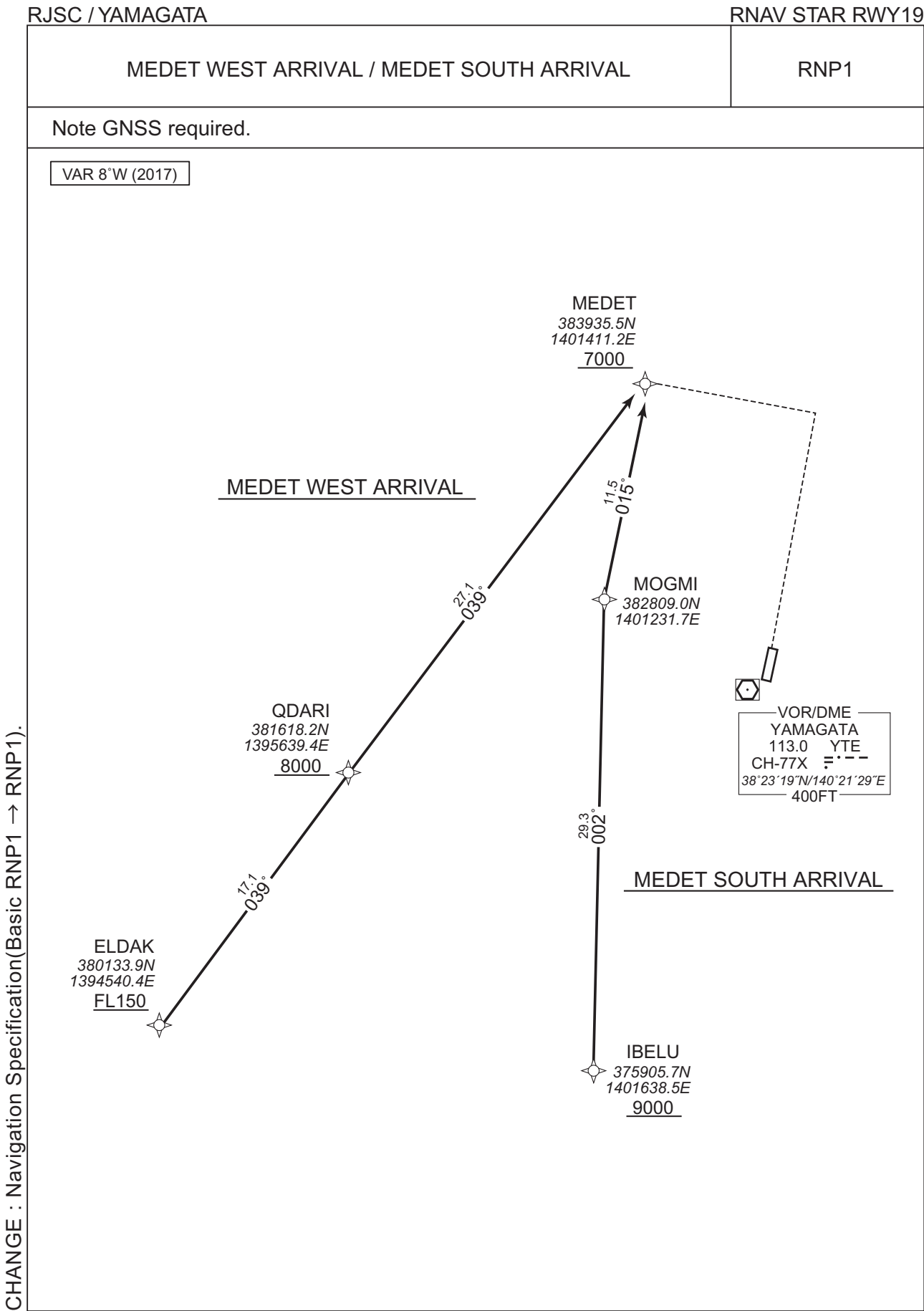
YOZAN NORTH ARRIVAL

From YTE, to IMONY, to YOZAN at or above 9000FT.

| Serial Number | Path Descriptor | Waypoint Identifier | Fly Over | Course °M(°T)  | Magnetic Variation | Distance (NM) | Turn Direction | Altitude (FT) | Speed (KIAS) | Vertical Angle | Navigation Specification |
|---------------|-----------------|---------------------|----------|----------------|--------------------|---------------|----------------|---------------|--------------|----------------|--------------------------|
| 001           | IF              | YTE                 | -        | -              | -8.2               | -             | -              | -             | -            | -              | RNP1                     |
| 002           | TF              | IMONY               | -        | 225<br>(216.5) | -8.2               | 15.1          | -              | -             | -250         | -              | RNP1                     |
| 003           | TF              | YOZAN               | -        | 195<br>(186.4) | -8.2               | 4.9           | -              | +9000         | -            | -              | RNP1                     |

CHANGE : Navigation Specification(Basic RNP1 → RNP1).

STANDARD ARRIVAL CHART - INSTRUMENT



STANDARD ARRIVAL CHART - INSTRUMENT

RJSC / YAMAGATA

RNAV STAR RWY19

MEDET WEST ARRIVAL

From ELDAK at or above FL150, to QDARI at or above 8000FT,  
to MEDET at or above 7000FT.

| Serial Number | Path Descriptor | Waypoint Identifier | Fly Over | Course °M(°T)  | Magnetic Variation | Distance (NM) | Turn Direction | Altitude (FT) | Speed (KIAS) | Vertical Angle | Navigation Specification |
|---------------|-----------------|---------------------|----------|----------------|--------------------|---------------|----------------|---------------|--------------|----------------|--------------------------|
| 001           | IF              | ELDAK               | -        | -              | -8.2               | -             | -              | +FL150        | -            | -              | RNP1                     |
| 002           | TF              | QDARI               | -        | 039<br>(030.3) | -8.2               | 17.1          | -              | +8000         | -            | -              | RNP1                     |
| 003           | TF              | MEDET               | -        | 039<br>(030.4) | -8.2               | 27.1          | -              | +7000         | -            | -              | RNP1                     |

MEDET SOUTH ARRIVAL

From IBELU at or above 9000FT, to MOGMI, to MEDET at or above 7000FT.

| Serial Number | Path Descriptor | Waypoint Identifier | Fly Over | Course °M(°T)  | Magnetic Variation | Distance (NM) | Turn Direction | Altitude (FT) | Speed (KIAS) | Vertical Angle | Navigation Specification |
|---------------|-----------------|---------------------|----------|----------------|--------------------|---------------|----------------|---------------|--------------|----------------|--------------------------|
| 001           | IF              | IBELU               | -        | -              | -8.2               | -             | -              | +9000         | -            | -              | RNP1                     |
| 002           | TF              | MOGMI               | -        | 002<br>(353.7) | -8.2               | 29.3          | -              | -             | -            | -              | RNP1                     |
| 003           | TF              | MEDET               | -        | 015<br>(006.5) | -8.2               | 11.5          | -              | +7000         | -            | -              | RNP1                     |

CHANGE : Navigation Specification(Basic RNP1 → RNP1).



STANDARD ARRIVAL CHART - INSTRUMENT

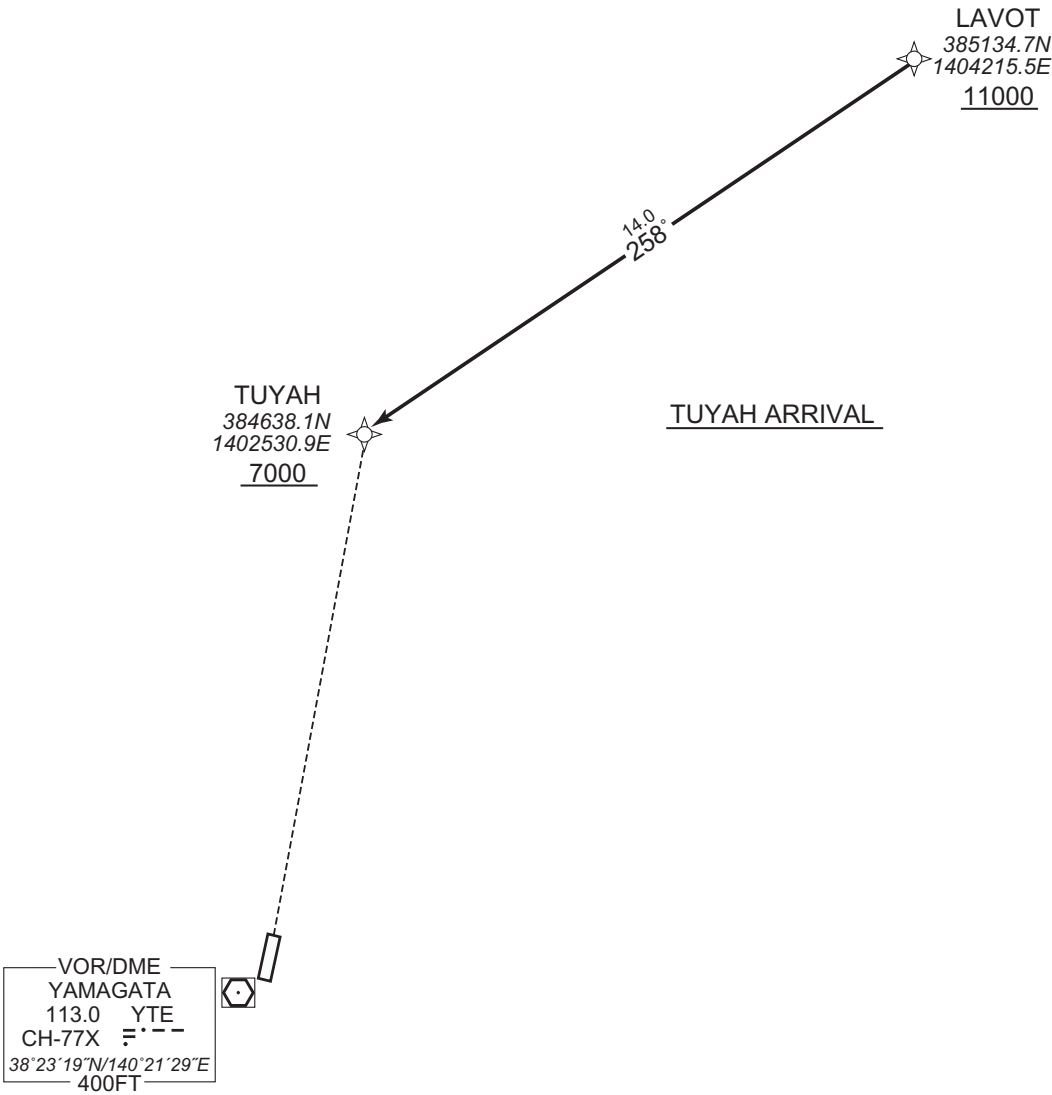
RJSC / YAMAGATA

RNAV STAR RWY19

| TUYAH ARRIVAL | RNP1 |
|---------------|------|
|---------------|------|

Note GNSS required.

VAR 8°W (2017)



TUYAH ARRIVAL

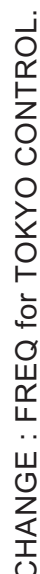
From LAVOT at or above 11000FT, to TUYAH at or above 7000FT.

| Serial Number | Path Descriptor | Waypoint Identifier | Fly Over | Course °M(°T) | Magnetic Variation | Distance (NM) | Turn Direction | Altitude (FT) | Speed (KIAS) | Vertical Angle | Navigation Specification |
|---------------|-----------------|---------------------|----------|---------------|--------------------|---------------|----------------|---------------|--------------|----------------|--------------------------|
| 001           | IF              | LAVOT               | —        | —             | -8.2               | —             | —              | +11000        | —            | —              | RNP1                     |
| 002           | TF              | TUYAH               | —        | 258 (249.3)   | -8.2               | 14.0          | —              | +7000         | —            | —              | RNP1                     |

CHANGE : Navigation Specification(Basic RNP1 → RNP1).

## RJSC / YAMAGATA

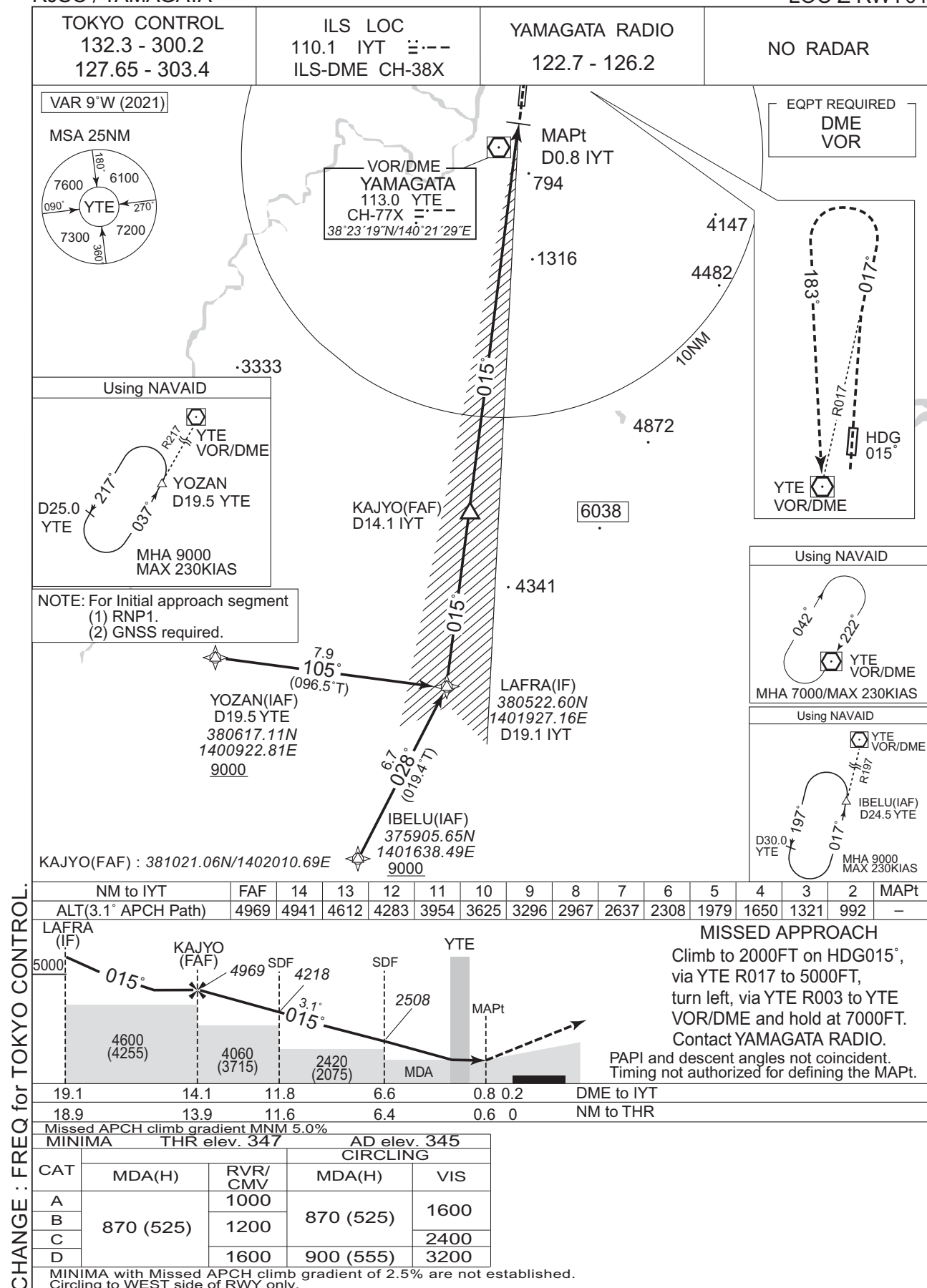
ILS Z RWY01



## INSTRUMENT APPROACH CHART

RJSC / YAMAGATA

LOC Z RWY01



## RJSC / YAMAGATA

|                                                         |                                                                                 |                                        |                 |
|---------------------------------------------------------|---------------------------------------------------------------------------------|----------------------------------------|-----------------|
| <b>TOKYO CONTROL</b><br>132.3 - 300.2<br>127.65 - 303.4 | <b>ILS LOC</b><br>110.1 IYT ---<br><b>ILS-GP</b> 334.4<br><b>ILS-DME</b> CH-38X | <b>YAMAGATA RADIO</b><br>122.7 - 126.2 | <b>NO RADAR</b> |
|---------------------------------------------------------|---------------------------------------------------------------------------------|----------------------------------------|-----------------|

VAR 9°W (2021)

**MSA 25NM**

**VOR/DME**  
**YAMAGATA**  
 113.0 YTE  
 CH-77X ---  
 38°23'19"N/140°21'29"E

Turn initiation within D17.0 YTE  
MAX Turning speed 200KIAS

EQPT REQUIRED  
**DME**  
**VOR**

KAJYO(FAF) : 381021.06N/1402010.69E

YTE  
VOR/DME

MHA 7000/MAX 230KIAS

Turn initiation within D17.0 YTE

209°

7000

KAJYO (FAF)

015°

5000

GP 3.0°

015°

RDH54

14.1

13.9

0.2

DME to IYT NM to THR

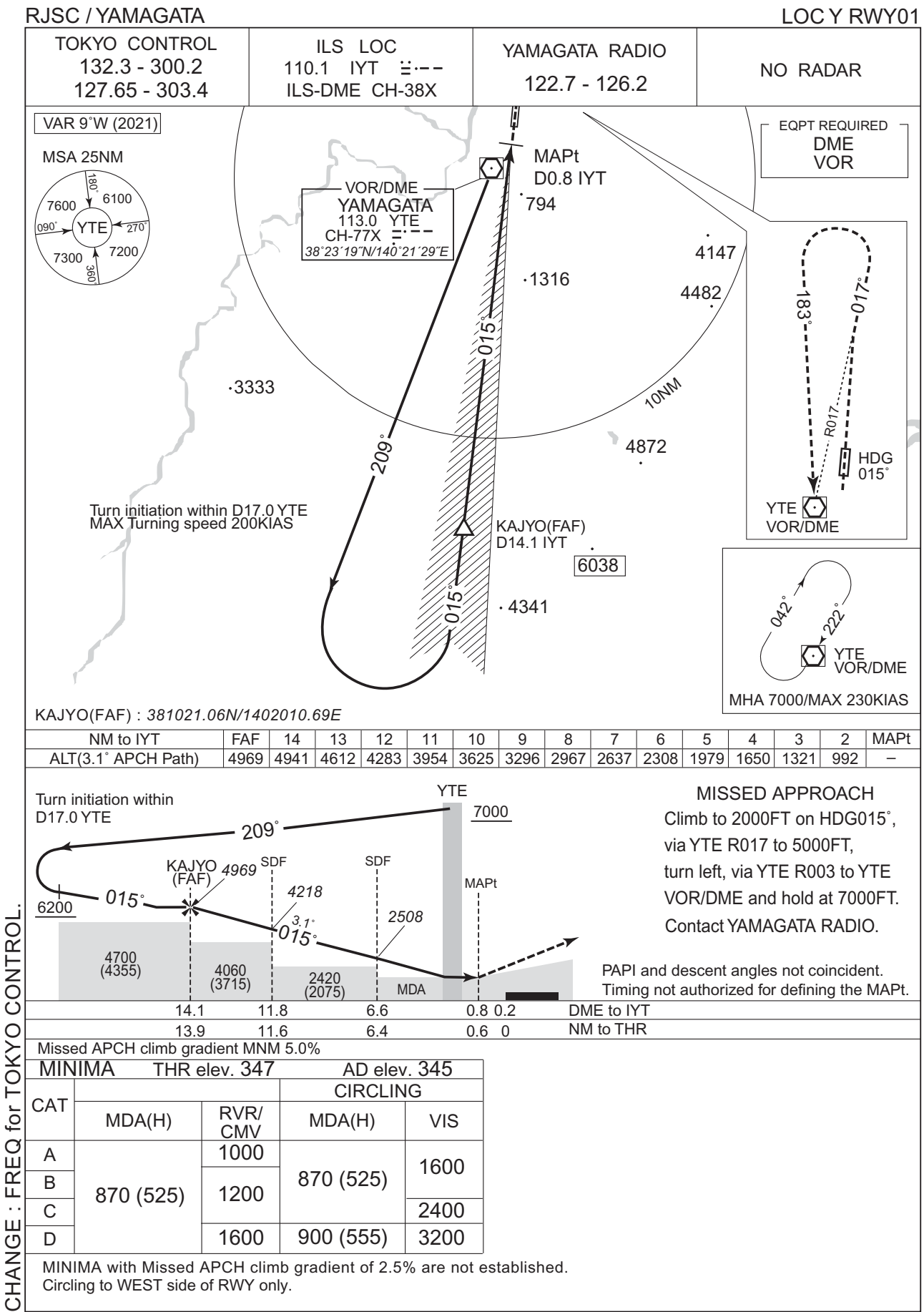
**MISSED APPROACH**  
 Climb to 2000FT on HDG015°,  
 via YTE R017 to 5000FT,  
 turn left, via YTE R003 to YTE  
 VOR/DME and hold at 7000FT.

Contact YAMAGATA RADIO.

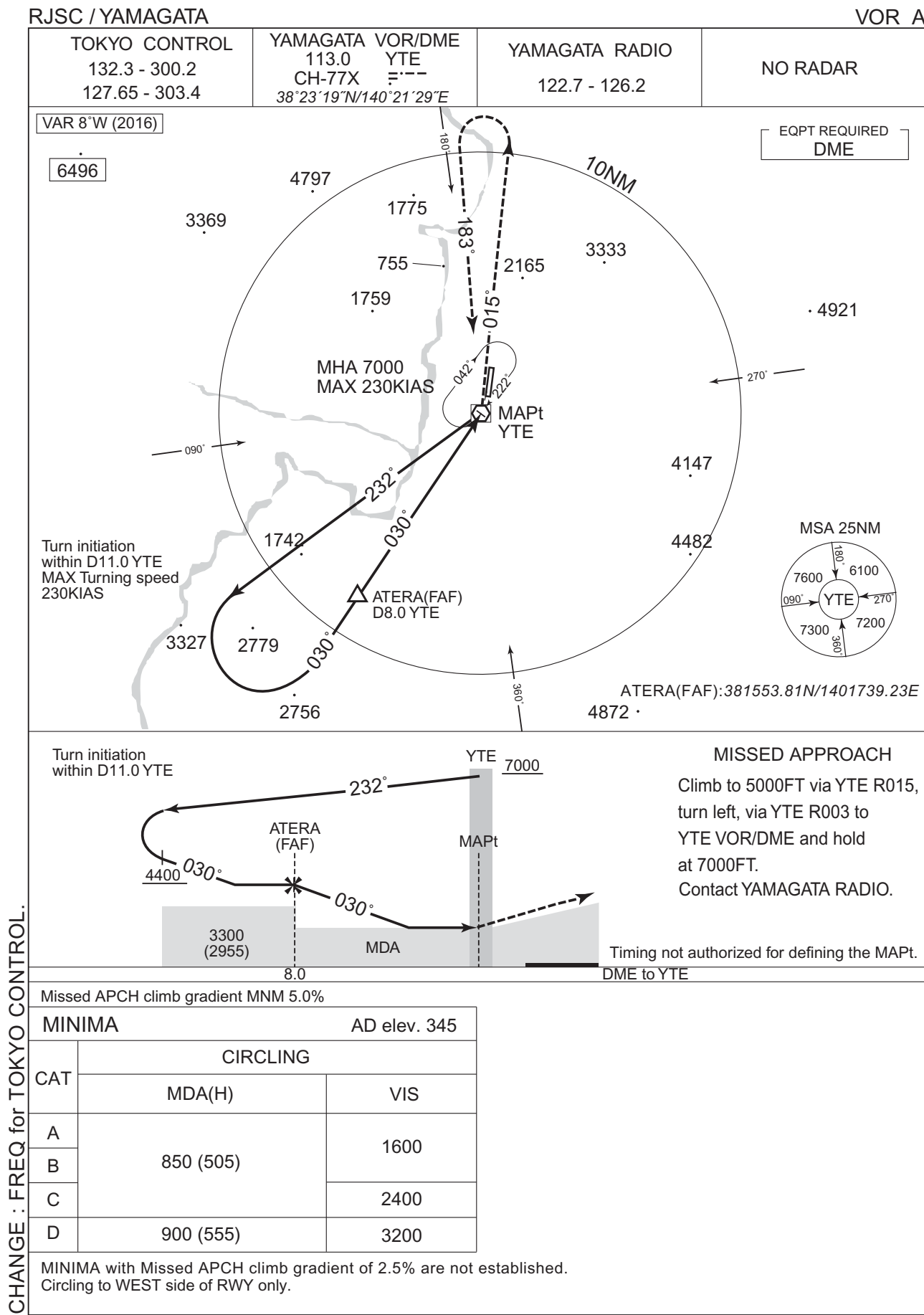
|                                     |           |               |              |      |
|-------------------------------------|-----------|---------------|--------------|------|
| Missed APCH climb gradient MNM 5.0% |           |               |              |      |
| MINIMA                              |           | THR elev. 347 | AD elev. 345 |      |
| CAT                                 | CAT I     |               | CIRCLING     |      |
|                                     | DA(H)     | RVR/<br>CMV   | MDA(H)       | VIS  |
| A                                   | 547 (200) | 550           | 790 (445)    | 1600 |
| B                                   |           |               | 800 (455)    |      |
| C                                   |           |               |              | 2400 |
| D                                   |           |               | 900 (555)    | 3200 |

MINIMA with Missed APCH climb gradient of 2.5% are not established.  
Circling to WEST side of RWY only.

INSTRUMENT APPROACH CHART



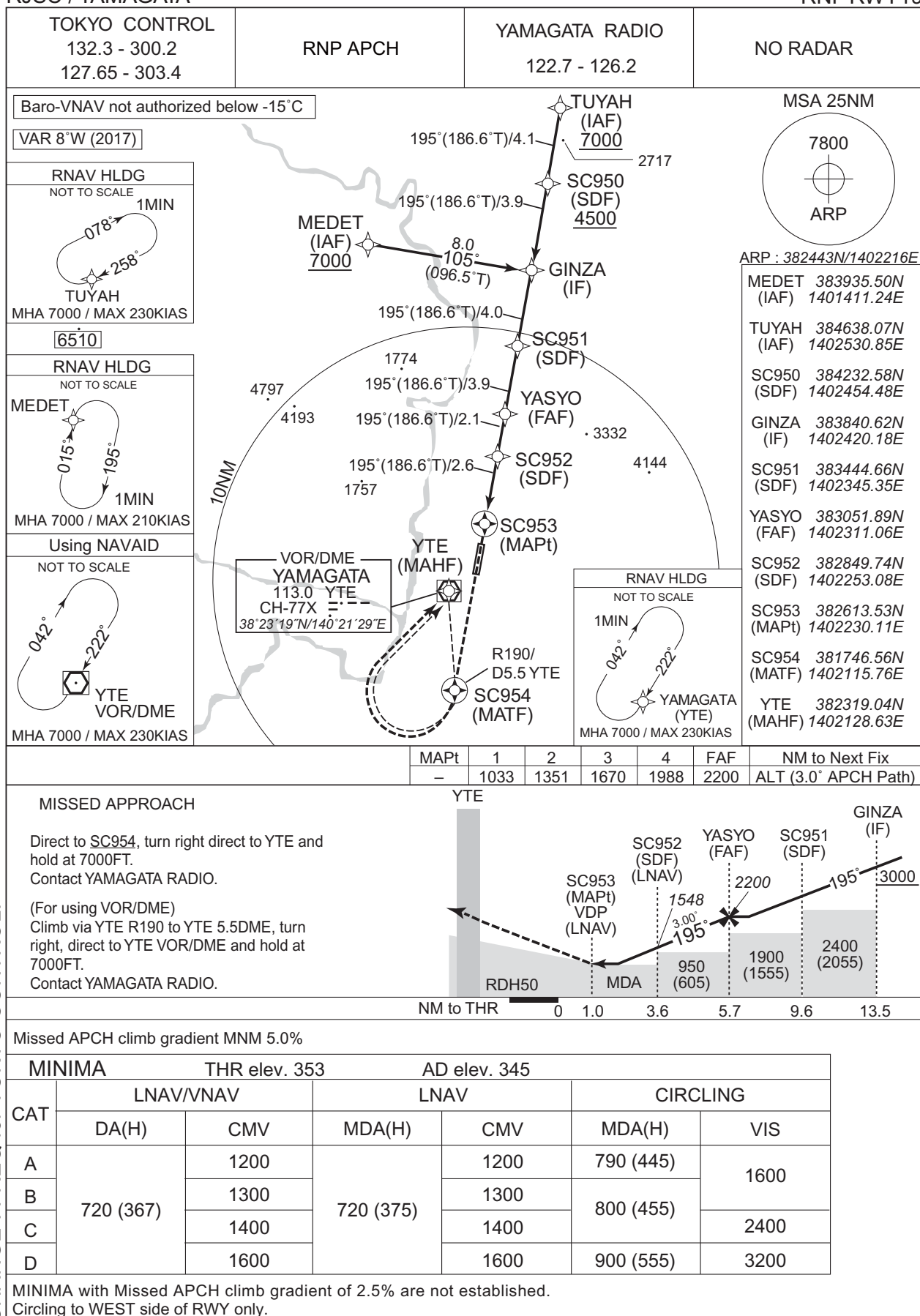
INSTRUMENT APPROACH CHART



## INSTRUMENT APPROACH CHART

RJSC / YAMAGATA

RNP RWY19

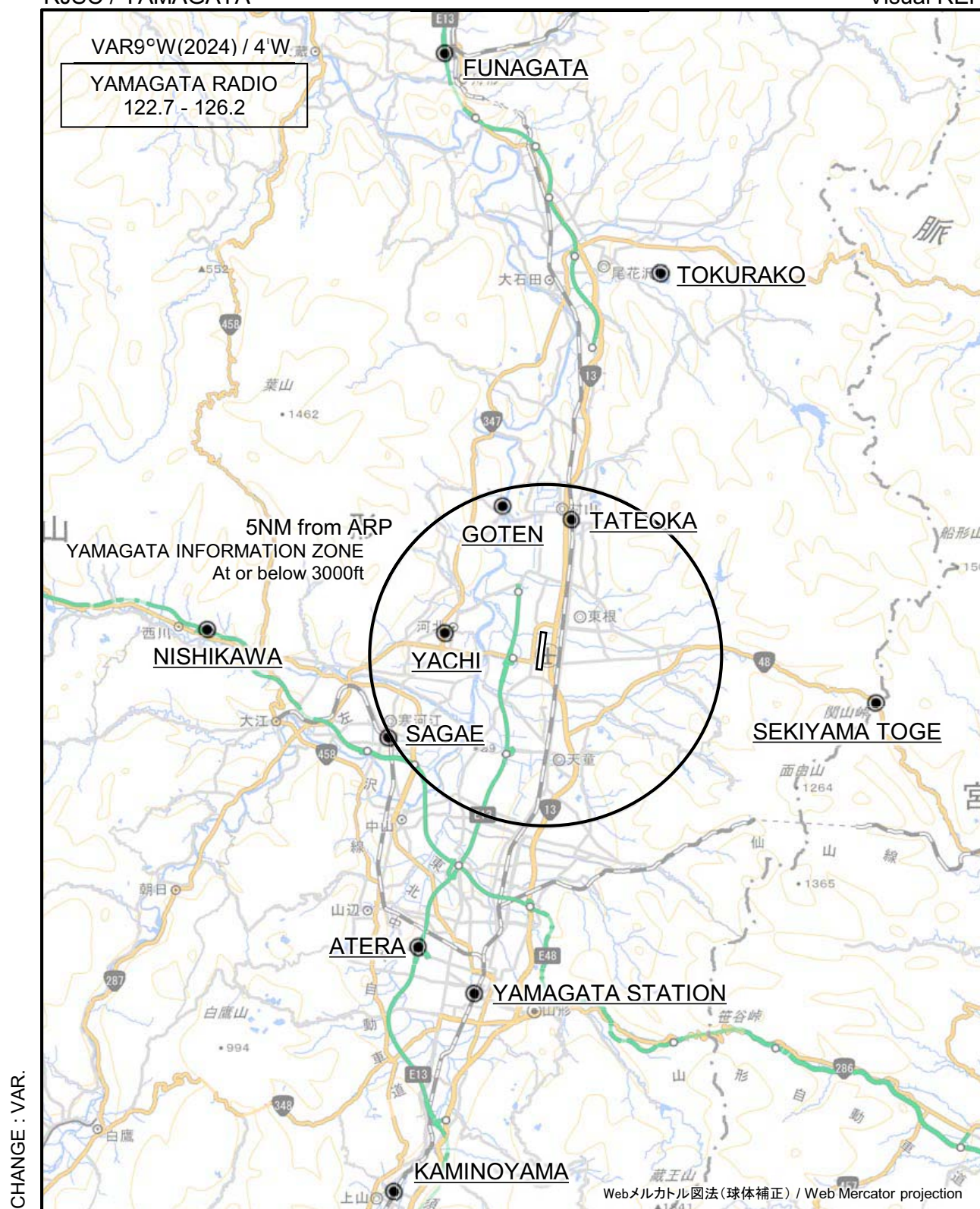


CHANGE : FREQ for TOKYO CONTROL.



RJSC / YAMAGATA

Visual REP



※図中に標高を示す数字がある場合、単位はメートル(m)である。The unit of measurement used to express elevation is meter(m).

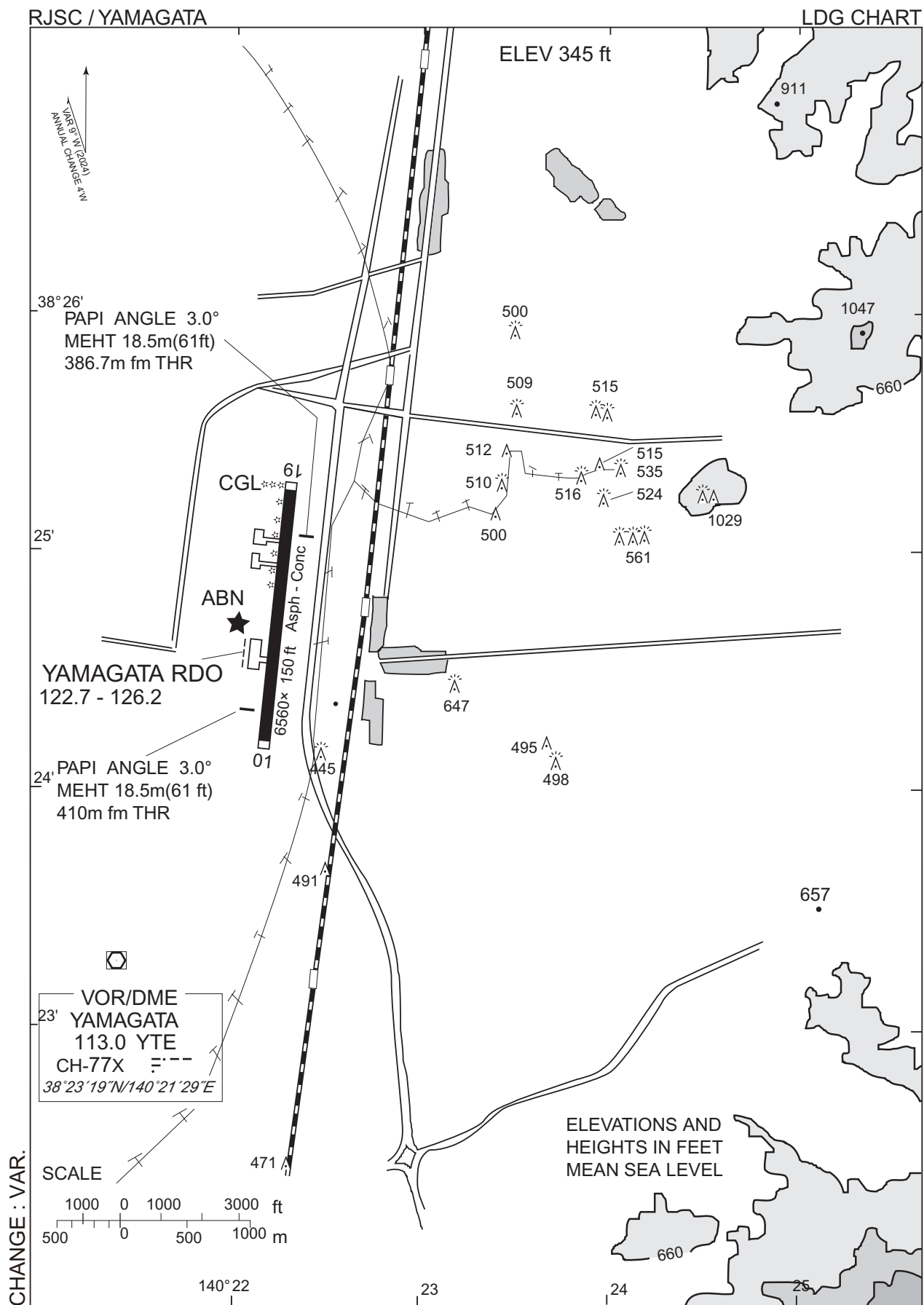


RJSC / YAMAGATA

Visual REP

| Call sign                    | BRG / DIST from ARP | Remarks                     |
|------------------------------|---------------------|-----------------------------|
| 舟形<br>Funagata               | 351°T / 17.7NM      | 舟形インターチェンジ<br>Interchange   |
| 徳良湖<br>Tokurako              | 017°T / 11.5NM      | 徳良湖<br>Lake                 |
| 基点<br>Goten                  | 344°T / 4.5NM       | 村山市民体育館<br>Gymnasium        |
| 楯岡<br>Tateoka                | 010°T / 4.0NM       | 村山駅(旧楯岡駅)<br>JR station     |
| 西川<br>Nishikawa              | 274°T / 9.8NM       | 西川インターチェンジ<br>Interchange   |
| 谷地<br>Yachi                  | 282°T / 3.0NM       | 谷地高校<br>School              |
| 関山峠<br>Sekiyama Toge         | 100°T / 9.5NM       | 関山トンネル<br>Tunnel            |
| 寒河江<br>Sagae                 | 242°T / 5.1NM       | 寒河江駅<br>JR station          |
| アテラ<br>Atera                 | 203°T / 9.3NM       | 山形中央インターチェンジ<br>Interchange |
| 山形ステーション<br>Yamagata Station | 192°T / 10.0NM      | 山形駅<br>JR station           |
| 上山<br>Kaminoyama             | 196°T / 16.2NM      | かみのやま温泉駅<br>JR station      |

CHANGE : Kaminoyama(Remarks).



RJSC / YAMAGATA

Minimum Vectoring Altitude CHART

